Introduction to Research and Identification of the Research Problem: A Rigorous Methodological Framework for Computer Science and Engineering

Academic Research Scholar
Department of Computer Science and Applications
Faculty of Science and Technology
Course: Research Methodology (BCA/MCA/B.Tech)
Email: scholar.researcher@academic.edu

***Abstract*—The formulation of an actionable, methodologically sound research problem represents the single most critical milestone in Computer Science and Engineering scholarship. Beginning undergraduate researchers frequently conflate broad technology domains with research problems, resulting in descriptive implementation exercises that lack scholarly contribution, falsifiable hypotheses, and peer-reviewed rigor. This paper establishes an authoritative, comprehensive pedagogical framework designed to instruct students in the systematic transition from literature exploration to empirical problem formulation. We formalize the epistemological foundations and taxonomy of research (spanning basic, applied, empirical, experimental, and mixed-methods paradigms); delineate the fifteen-stage scientific research progression; and provide a structured comparative framework distinguishing research areas, topics, problems, gaps, questions, and objectives. Furthermore, we articulate an adversarial keyword-driven protocol for mining explicit limitations and contradictions from peer-reviewed literature, formalize a ten-fold taxonomy of research gaps, and establish a six-stage problem statement architecture. Finally, we demonstrate the unbroken methodological chain from problem identification to experimental execution through an evidence-grounded Computer Science exemplar in dermatological deep learning on the HAM10000 benchmark, concluding with a ten-dimensional quality rubric for evaluating candidate research questions.**

***Index Terms*—Research Methodology, Research Problem Identification, Literature Gap Analysis, Research Questions, Research Objectives, Computer Science Education, Deep Learning, Empirical Software Engineering.**

I. Introduction

Research in Computer Science and Engineering (CSE) and Computer Applications (BCA/MCA) constitutes an empirical, systematic, and methodologically grounded investigation aimed at discovering novel insights, creating computational artifacts, or validating algorithmic hypotheses [1]. For undergraduate scholars and early-career researchers, the transition from classroom programming to rigorous scientific inquiry presents a profound paradigm shift: while engineering practice is traditionally concerned with the development of software artifacts satisfying known specifications, academic and scientific research is concerned with the systematic resolution of unknown problems, unverified assumptions, and unaddressed limitations in the state of the art [2], [3].

The most pervasive failure mode encountered by student researchers is the conflation of a *research topic* (e.g., "Skin Cancer Detection Using Deep Learning") with a *research problem* (e.g., "The severe performance degradation of convolutional neural networks on minority malignant lesions caused by acute class distribution skew and unweighted loss functions") [4]. Without a mathematically or experimentally falsifiable problem, research inquiries collapse into aimless implementation exercises lacking scholarly contribution.

This document serves as an exhaustive, authoritative pedagogical and methodological framework. It articulates the foundational philosophy of research, formalizes the taxonomy of research types, establishes the end-to-end scientific pipeline, presents a structured protocol for discovering research gaps in literature, details the 6-stage problem statement architecture, and demonstrates the translation of research problems into testable questions, measurable objectives, and empirical methodologies.

II. Understanding Research

A. Formal Definitions and Theoretical Grounding

The term *research* derives etymologically from the Middle French *recherchier*, signifying an intensive, exhaustive search. In formal scientific literature, research is systematically defined across multiple methodological traditions:

- **Creswell's Definition:** Creswell and Creswell [2] define research as a cyclical process of steps used to collect and analyze information to increase our understanding of a topic or issue, consisting of posing a question, collecting data to answer the question, and presenting an answer.
- **Kothari's Definition:** Kothari and Garg [4] formalize research as a scientific and systematic search for pertinent information on a specific topic, characterized by a systematized effort to gain new knowledge through controlled, verifiable, and empirical techniques.
- **OECD Frascati Definition:** The OECD Frascati Manual [5] defines research and experimental development (R&D) as creative and systematic work undertaken in order to increase the stock of knowledge—including

knowledge of humankind, culture, and society—and to devise new applications of available knowledge. To qualify as R&D, an activity must satisfy five core criteria: novel, creative, uncertain, systematic, and transferable/reproducible.

B. Why Research is Conducted

Research is undertaken to serve several distinct epistemological and pragmatic functions:

1) **Discovery of New Knowledge:** Uncovering fundamental computational principles, mathematical properties of algorithms, or biological phenomena.
2) **Resolution of Unsolved Problems:** Overcoming computational bottlenecks, algorithmic complexity limits, security vulnerabilities, or classification failures under non-ideal data distributions.
3) **Resolution of Contradictory Evidence:** Systematically arbitrating between conflicting published findings through controlled benchmarking and statistical hypothesis testing.
4) **Validation and Generalization:** Establishing the boundary conditions, transferability, and robustness of existing models across unseen domains, hardware architectures, and operating contexts.
5) **Engineering Artifact Innovation:** Designing novel systems, compilers, architectures, or protocols that provide measurable improvements over existing baselines.

C. Characteristics of Good Research

Authoritative methodology literature identifies seven indispensable characteristics of sound scientific research [3], [4]:

1) **Systematic:** Structured according to predefined, logically ordered procedures. Research does not proceed by erratic trial-and-error; it follows a rigorous protocol.
2) **Controlled:** In empirical investigations, confounding variables are held constant or accounted for so that causal relationships between independent and dependent variables can be isolated.
3) **Empirical:** Grounded in observable, measurable, verifiable reality rather than mere speculation or unsupported subjective assertion.
4) **Critical and Skeptical:** Scrutinized through rigorous verification, error estimation, and vulnerability to peer review and falsification.
5) **Rigorous and Precise:** Operating under exact mathematical formulations, explicit definitions, and standardized evaluation protocols.
6) **Valid and Verifiable:** Generating results whose internal validity (freedom from experimental bias) and external validity (generalizability) can be independently evaluated.
7) **Replicable and Reproducible:** Documented with sufficient methodological transparency (including code, datasets, hyperparameter choices, and hardware environments) that independent scholars can duplicate the experimental setup and achieve equivalent outcomes.

D. Taxonomy of Research Paradigms in Computing

Research is categorized across distinct epistemological domains:

- **Scientific Research:** Focuses on discovering underlying natural or computational laws through observation, hypothesis formulation, experimentation, and induction/deduction.
- **Academic Research:** Conducted within scholarly institutions to advance theoretical understanding, critique existing literature, and train future scholars under peer-reviewed standards.
- **Engineering Research:** Focuses on systematic design, synthesis, optimization, and evaluation of technical systems, balancing trade-offs between performance, resource consumption, cost, and safety.
- **Computer Science Research:** Spans theoretical computer science (computational complexity, automata, formal verification), systems engineering (operating systems, distributed architectures, databases), and empirical computing (machine learning, human-computer interaction, empirical software engineering) [1]. As Mary Shaw [1] formalizes, CS research asks questions about methods, models, tools, and empirical behaviors, evaluating answers through mathematical proofs, qualitative analysis, or controlled benchmarks.
- **Basic / Fundamental Research:** Seeks fundamental theoretical insight and expansion of universal knowledge without immediate commercial or utilitarian application (e.g., proving computational complexity bounds of NP-hard problems).
- **Applied Research:** Directed primarily toward solving specific, practical, real-world problems by applying known theoretical principles to concrete technical challenges (e.g., developing edge-deployable deep learning algorithms for melanoma triage in teledermatology) [5].

III. Types of Research

A critical epistemological realization for computing students is that *research classifications are not mutually exclusive*. Rather, research paradigms represent orthogonal dimensions: a single study may simultaneously be **applied** in purpose, **quantitative** in data representation, **experimental** in strategy, and **empirical** in epistemological foundation.

Table I synthesizes the fourteen major classifications of research, defining their purpose, typical methodologies, advantages, limitations, and direct Computer Science implementations.

IV. The Research Process

Scientific research in Computer Science is a structured, cyclical, and self-correcting progression. The linear depiction in Fig. 1 represents the logical architecture of research design, though researchers frequently iterate backwards when empirical roadblocks occur.

The specific operational activities executed at each stage comprise:

THE SYSTEMATIC RESEARCH PIPELINE:

1) **Research Area** (Broad domain selection)
2) **Research Topic** (Focused thematic scoping)
3) **Literature Search** (Systematic query-driven indexing)
4) **Literature Review** (Critical synthesis and gap extraction)
5) **Research Problem** (Formulating the core unresolved dilemma)
6) **Research Gap** (Documenting the precise missing knowledge)
7) **Research Question** (Formulating testable, falsifiable inquiries)
8) **Research Objectives** (Defining measurable engineering deliverables)
9) **Research Methodology** (Designing the experimental architecture)
10) **Data Collection / Dataset** (Curating and partitioning data)
11) **Experiment / Analysis** (Controlled model training and ablation)
12) **Results** (Quantifying empirical performance)
13) **Discussion** (Synthesizing insights, threats to validity)
14) **Conclusion** (Summarizing key findings and impact)
15) **Future Work** (Identifying next-generation scientific horizons)

Fig. 1. The systematic research progression in Computer Science and Engineering.

1) **Research Area:** Identification of a broad computing domain aligned with academic relevance and societal significance (e.g., Computer Vision, Artificial Intelligence).
2) **Research Topic:** Scoping the area down to a concrete application domain or computational subfield (e.g., Automated Skin Lesion Classification Using Deep Learning).
3) **Literature Search:** Executing systematic bibliographic queries across authoritative indexing databases (IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink) using boolean operators and inclusion/exclusion protocols [7].
4) **Literature Review:** Critically analyzing, synthesizing, and tabulating existing approaches into a structured literature matrix, rather than writing isolated summaries.
5) **Research Problem:** Abstracting the core unresolved obstacle, contradiction, or limitation demonstrated in prior work into a formal academic problem.
6) **Research Gap:** Pinpointing the exact missing piece of evidence, unaddressed technical challenge, or invalid assumption in the literature that warrants investigation.
7) **Research Question (RQ):** Converting the research gap into sharp, falsifiable, experimentally investigable questions that guide the inquiry.
8) **Research Objectives:** Constructing measurable, action-oriented engineering and analytical commitments that directly answer the RQs.
9) **Research Methodology:** Defining the formal algorithms, network architectures, baseline configurations, independent/dependent variables, and ablation protocols.
10) **Data Collection / Dataset:** Selecting, verifying, cleaning, deduplicating, and partitioning representative benchmark data under strict leak-free splits.
11) **Experimentation / Implementation:** Executing controlled algorithmic implementations under identical computational environments, tracking hyperparameters and seeds.
12) **Results:** Tabulating and plotting quantitative metrics (e.g., Precision, Recall, Macro F1, AUC-ROC, Inference Latency) with statistical dispersion measures (standard deviation, confidence intervals).
13) **Discussion:** Interpreting the results against prior literature, explaining *why* specific behaviors emerged, analyzing trade-offs, and documenting threats to validity.
14) **Conclusion:** Synthesizing the core findings, asserting the primary scholarly contribution, and detailing the boundary conditions of the work.
15) **Future Work:** Articulating open questions, prospective technical extensions, and unexplored avenues exposed by the current study's limitations.

V. Research Topic vs. Research Problem

A foundational milestone in research methodology education is distinguishing between broad conceptual labels and actionable research formulations. Beginning students frequently present an overarching topic as their problem, stating: "My research problem is skin cancer detection using AI." This is invalid: skin cancer detection is an engineering *topic* or *application domain*, not an academic *problem*.

Table II establishes the definitive taxonomy of research concepts, providing conceptual definitions, functional roles, and an end-to-end Computer Science exemplar.

VI. Identification of a Research Problem

Where do valid, publishable research problems originate? In computing, valid problems do not emerge from isolated guesswork; they are mined systematically from the friction points of technology and literature [11].

A. Primary Sources of Research Problems

1) **Critical Literature Gaps:** Direct analysis of peer-reviewed literature revealing unexamined phenomena, unaddressed limitations, or overlooked variables.
2) **Contradictory Empirical Findings:** Disagreements across published studies. For instance, Paper A concludes that transfer learning fine-tuning of all layers is superior, while Paper B asserts it induces catastrophic forgetting on medical images. Such contradictions demand systematic comparative arbitration.
3) **Methodological Limitations:** Studies relying on invalid shortcuts, such as training and evaluating on augmented data before splitting (data leakage), evaluating multi-class imbalance using only raw accuracy, or omitting cross-validation.
4) **Computational Bottlenecks:** Algorithms that achieve high accuracy but require excessive memory, energy, or latency, preventing deployment in constrained edge, IoT, or real-time clinical environments.
5) **Dataset Skew and Lack of Diversity:** Models trained exclusively on homogeneous populations (e.g., fair skin types in Western dermatology archives) failing to generalize to diverse demographic cohorts or differing camera hardware.

6) **Domain Generalization Failure:** High-performing models suffering dramatic accuracy collapse ($> 10\%$) when tested on external clinic benchmarks due to domain shift, varying illumination, or background artifacts.
7) **Black-Box Interpretability Deficits:** High accuracy coupled with complete absence of explainability, preventing clinical adoption where practitioners must verify the diagnostic rationale behind algorithmic predictions.
8) **Replication and Verification Inquiries:** Re-implementing high-profile algorithms under strict, leak-free, public benchmark conditions to verify whether reported performance reproduces faithfully.

VII. How to Read a Paper to Find a Research Problem

Undergraduate students often read papers passively as textbooks, attempting to memorize every formula. To extract research problems, students must transition to *adversarial and critical reading*, interrogating the text for structural weaknesses and explicit author confessions.

A. The Keyword-Triggered Gap-Hunting Protocol

Academic authors are ethically bound to disclose the boundary conditions of their work. Consequently, author-stated limitations cluster around specific rhetorical marker words. Table III formalizes the critical reading triggers that expose candidate research problems.

B. The Critical Anti-Fallacy Rule

> **METHODOLOGICAL PRINCIPLE:** *Do not assume that every sentence containing "however" represents an actionable research gap.*

In academic writing, the word "however" frequently serves purely rhetorical or pedagogical functions:

- **Stylistic Contrast:** "CNNs are effective; however, they require GPU acceleration." (A well-known computational property, not a novel research gap).
- **Scope Delimitation:** "We acknowledge the existence of segmentation methods; however, our study focuses strictly on classification." (An authorial scoping decision, not necessarily an unsolved gap).
- **Resolved Contrast:** "Historically, training was slow; however, our proposed optimizer achieves convergence in 5 epochs." (A problem solved within the paper itself).

To qualify as an authentic research problem, an author's stated limitation must be verified as **unresolved**, **technically significant**, **experimentally tractable**, and **systematically reproducible**.

VIII. Typology of Research Gaps

A *research gap* is a demonstrable absence of knowledge, evaluation, methodology, or empirical evidence in the established scientific literature [11]. Research gaps are categorized into ten distinct types:

1) **Knowledge Gap:** A void in conceptual understanding or theoretical modeling of a computational process.
   - *CS Example:* Lack of mathematical characterization regarding how loss landscapes change when fine-tuning Vision Transformers versus CNNs on small medical datasets.
   - *Identification:* Scrutinizing theoretical frameworks and literature reviews for unmodeled phenomena.
   - *Conversion to RQ:* "How do the Hessian eigenvalue spectra of fine-tuned ViTs compare to CNNs under small sample regimes?"
2) **Methodological Gap:** A limitation in the algorithms, architectures, optimization techniques, or preprocessing pipelines previously deployed.
   - *CS Example:* Prior studies relied exclusively on simple random oversampling to address lesion class imbalance, inducing severe memorization and overfitting on duplicate samples [9].
   - *Identification:* Examining methodology and implementation sections of existing surveys.
   - *Conversion to RQ:* "To what extent does latent feature-space augmentation reduce validation overfitting compared to naive image-space duplication in multi-class skin lesion classification?"
3) **Empirical Gap:** A lack of verifiable experimental data or empirical validation under controlled conditions.
   - *CS Example:* No published study has systematically benchmarked the inference throughput, energy consumption, and macro-F1 score of MobileNetV3 on edge hardware against desktop GPUs for dermatological triage.
   - *Identification:* Reviewing empirical result tables for missing hardware benchmarks.
   - *Conversion to RQ:* "What is the empirical trade-off between energy consumption, inference latency, and diagnostic sensitivity when executing quantized CNNs on ARM-based edge platforms?"
4) **Theoretical Gap:** A contradiction or absence in the theoretical principles that explain algorithmic convergence or behavior.
   - *CS Example:* Lack of theoretical justification for why self-attention mechanisms generalize better over irregular skin lesion boundaries than fixed-kernel convolutional operators.
   - *Identification:* Searching discussions where authors describe results as purely empirical observations without theoretical grounding.
   - *Conversion to RQ:* "Can the receptive field expansion of self-attention explain the variance in boundary segmentation performance across non-convex melanoma lesions?"
5) **Data Gap:** Absence of adequate, annotated, or balanced datasets required to evaluate a computational task.
   - *CS Example:* Benchmark archives lack dermoscopic images depicting rare skin diseases on dark skin phenotypes (Fitzpatrick skin types V and VI) [8].

- *Identification:* Auditing dataset demographic tables and sample distribution disclosures.
- *Conversion to RQ:* "How does the diagnostic error rate of deep learning classifiers vary across distinct Fitzpatrick skin phototype cohorts?"

6) **Population / Context Gap:** A phenomenon evaluated in one operational context that has never been validated in another.
   - *CS Example:* Computer vision models trained on high-resolution clinical dermatoscope images deployed on consumer-grade mobile smartphone cameras.
   - *Identification:* Comparing the training data acquisition environment with real-world target operational settings.
   - *Conversion to RQ:* "How severely does image artifact noise from low-cost smartphone lenses degrade the sensitivity of dermoscopic deep classifiers?"
7) **Contradictory Evidence Gap:** Published literature presents mutually incompatible findings regarding the same technique.
   - *CS Example:* Study A reports that transfer learning from ImageNet outperforms training from scratch on dermoscopy, while Study B asserts ImageNet features induce negative transfer due to domain distance.
   - *Identification:* Comparing reported outcomes across systematic literature matrices.
   - *Conversion to RQ:* "Under what dataset size thresholds does ImageNet feature pretraining provide statistically superior convergence over domain-specific pretraining?"
8) **Technology Gap:** Emergence of new computational architectures, hardware accelerators, or libraries that have not yet been evaluated on established problems.
   - *CS Example:* State-of-the-art Mamba state-space models have not been evaluated against established Transformers and CNNs on skin lesion multi-class classification.
   - *Identification:* Tracking recent fundamental architecture breakthroughs and identifying application domains where they remain untried.
   - *Conversion to RQ:* "How does the linear-time sequence modeling of Vision Mamba compare with quadratic-time Vision Transformers in multi-class skin lesion classification accuracy and GPU memory scaling?"
9) **Evaluation Gap:** Flawed, incomplete, or unstandardized evaluation metrics in prior studies.
   - *CS Example:* Numerous prior works report only raw overall accuracy on highly imbalanced datasets (HAM10000), concealing 0% sensitivity on life-threatening rare classes.
   - *Identification:* Examining performance metrics in literature and identifying the omission of Balanced Accuracy, Macro-F1, or Per-Class Recall.
   - *Conversion to RQ:* "How does model ranking change when evaluating skin cancer classifiers using Macro-Averaged Geometric Mean Recall rather than standard overall accuracy?"
10) **Generalization Gap:** Performance collapse observed when a model trained on one benchmark dataset is evaluated on an external, cross-center dataset without retraining.
   - *CS Example:* A model achieving 96% accuracy on HAM10000 plunges to 78% accuracy when evaluated on the MSK or BCN20000 dataset due to sensor and illumination shift.
   - *Identification:* Scrutinizing papers for the presence or absence of out-of-distribution external hospital testing.
   - *Conversion to RQ:* "To what extent can adaptive test-time normalization mitigate cross-center domain generalization degradation in dermoscopic classification?"

IX. Structuring the Formal Problem Statement

A problem statement is not a rambling narrative; it is an engineered argument comprising six mandatory structural components [1], [2]:

1) **Context:** The foundational background establishing why the general domain is important.
2) **Existing Situation:** The current state of technical adoption and mainstream algorithmic approaches.
3) **Existing Limitation:** The precise technical failure mode, bottleneck, or drawback in current methods.
4) **Consequence / Impact:** The real-world or computational cost incurred if this limitation remains unaddressed.
5) **Research Gap:** The documented void in literature where a solution has not been established.
6) **Need for Investigation:** The explicit call to action detailing what must be scientifically investigated.

A. Exemplar Computer Science Problem Statement

(Context) *Automated dermatological computer vision systems offer significant potential for early, non-invasive triage of malignant skin lesions.* ***(Existing Situation)*** *Contemporary state-of-the-art approaches routinely employ deep convolutional neural networks (e.g., ResNet, DenseNet, EfficientNet) and report overall classification accuracies exceeding 90% on standard benchmark datasets such as HAM10000.* ***(Existing Limitation)*** *However, standard public benchmarks exhibit severe class distribution skews, where common benign lesions (melanocytic nevi) constitute over 67% of images, while critical minority malignancies (such as dermatofibroma and vascular lesions) comprise less than 2% of samples. Under standard cross-entropy*

loss functions, gradient descent is heavily dominated by majority-class errors. ***(Consequence)*** *Consequently, deployed models exhibit alarmingly low diagnostic recall ($< 40\%$) on rare life-threatening lesions despite high aggregate accuracy, posing grave clinical risks of false-negative triage.* ***(Research Gap)*** *While heuristic oversampling techniques have been proposed, they frequently induce severe validation memorization, and the comparative efficacy of loss-reweighting schemes (such as focal loss and class-balanced loss) across diverse lightweight CNN backbones under strictly identical, leak-free preprocessing conditions remains unestablished in literature [9].* ***(Need for Investigation)*** *Therefore, a controlled empirical investigation is rigorously required to evaluate class-balanced loss formulations against standardized baselines, establishing an evidence-grounded framework for minority-class diagnostic sensitivity in resource-constrained dermatological environments.*

X. The 10-Step Problem Formulation Framework

To guide undergraduate scholars systematically from broad interest to experimental execution, this study provides the operational 10-step formulation framework illustrated below:

1) **Step 1: Select Broad Area:** Identify an established field of computing (e.g., Computer Vision and Deep Learning).
2) **Step 2: Narrow to Specific Topic:** Isolate a specialized problem domain (e.g., Multi-Class Skin Lesion Classification on Dermoscopic Images).
3) **Step 3: Search Existing Literature:** Query verified digital libraries (IEEE, ACM, Springer, Elsevier) and harvest 25–35 authoritative recent papers.
4) **Step 4: Identify Recurring Limitations:** Tabulate explicit author limitations regarding dataset skew, computational cost, and validation gaps.
5) **Step 5: Compare Existing Approaches:** Synthesize architectures, preprocessing pipelines, and evaluation metrics in a literature matrix.
6) **Step 6: Identify an Unresolved Issue:** Isolate a specific, tractable friction point supported across multiple papers (e.g., minority class recall failure under class imbalance).
7) **Step 7: Formulate the Research Problem:** Draft a 6-stage problem statement articulating context, reality, limitation, consequence, gap, and necessity.
8) **Step 8: Convert Problem into Research Questions:** Formulate sharp, focused, falsifiable inquiries containing clear independent and dependent variables.
9) **Step 9: Convert Research Questions into Objectives:** Draft measurable action statements using Bloom-aligned engineering verbs.
10) **Step 10: Select Methodology to Answer Questions:** Design the experimental protocol, datasets, models, loss functions, and statistical metrics capable of producing empirical evidence.

XI. Formulation of Research Questions

The Research Question (RQ) is the methodological engine of an academic investigation. It narrows the research problem down to an exact, interrogative inquiry that can be resolved through scientific observation, mathematical modeling, or controlled empirical experimentation [1], [2].

A. Criteria for High-Quality Research Questions

To qualify as methodologically robust in Computer Science, a research question must satisfy eight rigorous criteria:

1) **Clear and Unambiguous:** Phrased in precise technical terminology with well-defined semantics, leaving no doubt regarding what is being asked.
2) **Specific and Bounded:** Strictly delineated by clear operational boundaries (e.g., specific algorithms, datasets, performance constraints, and operating environments).
3) **Focused on Independent and Dependent Variables:** Explicitly identifies what is being manipulated or varied (independent variable) and what is being measured or observed (dependent variable).
4) **Empirically Researchable and Falsifiable:** Capable of being answered through real-world data collection, code execution, benchmark evaluation, or mathematical proof.
5) **Measurable via Standardized Metrics:** Yields outcomes that can be quantified using recognized computing metrics (e.g., F1-score, latency, memory footprint, ROC-AUC) rather than subjective impressions.
6) **Technically Feasible:** Solvable within the constraints of available compute resources, hardware accelerators, open-source datasets, and project timelines.
7) **Scholarly Relevant:** Meaningfully connected to the established state of the art, addressing a documented friction point or technical limitation.
8) **Directly Anchored in a Research Gap:** Designed specifically to bridge an identified void in peer-reviewed scientific literature.

B. Taxonomy of Research Question Forms in Computing

Research questions assume distinct morphological structures depending on the epistemological goal of the inquiry:

1) **Descriptive Research Questions:** Seek to systematically characterize the properties, distributions, or current operational baselines of a computing system or dataset.
   - *Structure:* "What is the distribution/prevalence of X in Y?"
   - *CS Example:* What is the distribution of lesion categories, image resolutions, and demographic metadata across open-access dermatological benchmarks including HAM10000 and ISIC 2019?
2) **Comparative Research Questions:** Contrast two or more algorithms, architectures, frameworks, or configurations under strictly identical conditions.
   - *Structure:* "How does architecture A compare with architecture B regarding metric M under condition C?"

- *CS Example:* How does EfficientNet-B0 compare with ResNet-50 and MobileNetV2 in top-1 macro-accuracy and GPU inference latency when classifying multi-class dermoscopic lesions on HAM10000 under identical preprocessing splits?

3) **Relational Research Questions:** Investigate the mathematical or statistical association between two continuous or categorical system variables.
   - *Structure:* "What is the relationship between variable X and variable Y in system Z?"
   - *CS Example:* What is the statistical correlation between deep neural network parameter depth and the magnitude of classification performance degradation observed under out-of-distribution domain shift?
4) **Causal Research Questions:** Determine whether manipulating a specific independent variable directly produces a measurable change in an outcome variable.
   - *Structure:* "Does the application of intervention X cause a statistically significant change in outcome Y?"
   - *CS Example:* Does replacing standard categorical cross-entropy loss with class-balanced focal loss cause a statistically significant increase in the recall of rare melanoma lesions in imbalanced training regimes?
5) **Evaluative Research Questions:** Assess the practical efficacy, viability, or trade-offs of a newly engineered system, tool, or protocol against established industrial or academic standards.
   - *Structure:* "To what extent does system S satisfy operational requirement R in deployment environment E?"
   - *CS Example:* To what extent can an 8-bit quantized convolutional neural network maintain clinically acceptable diagnostic sensitivity ($> 85\%$) while executing within a 100 ms inference latency budget on an ARM Cortex-A72 edge processor?
6) **Exploratory Research Questions:** Probe an emerging, uncharacterized problem space to generate novel hypotheses or map preliminary design spaces.
   - *Structure:* "How does system S behave when subjected to unprecedented condition C?"
   - *CS Example:* How do Vision-Language multimodal foundation models interpret out-of-vocabulary dermatological artifact noise when generating zero-shot diagnostic differential rationales?

XII. Good vs. Bad Research Questions

Table IV presents a critical comparative evaluation contrasting flawed research questions commonly submitted by novice scholars against rigorous, methodologically validated alternatives.

XIII. Formulation of Research Objectives

A Research Objective is a concrete, operationalized declaration of the actions the researcher will undertake to answer the research questions [4]. While the Research Question poses the scientific inquiry, the Research Objective defines the procedural commitment.

A. General vs. Specific Objectives

- **General Objective (Aim):** An overarching statement summarizing the primary ultimate goal of the investigation (e.g., "To develop, evaluate, and validate a computationally efficient, class-balanced deep transfer learning framework for robust multi-class skin lesion classification on public dermoscopic benchmarks").
- **Specific Objectives:** A logically sequenced series of operational milestones that collectively achieve the general objective and systematically answer each individual research question.

B. The Measurable Action Verb Hierarchy

To maintain academic rigor, research objectives must be phrased using precise, measurable action verbs derived from Bloom's Revised Taxonomy for Engineering and Scientific Domains [3]. Table V contrasts approved empirical verbs against unacceptable, non-measurable expressions.

C. Mapping Research Questions to Research Objectives

Every research question must have a corresponding, explicitly paired research objective. Table VI illustrates this direct methodological alignment.

XIV. Comprehensive Computer Science Exemplar: Deep Learning for Skin Lesion Diagnosis

To demonstrate the unbroken methodological chain from problem identification to experimental execution, this section details a complete, evidence-grounded exemplar within Computer Vision and Artificial Intelligence.

A. Step 1: Research Area

Artificial Intelligence, Machine Learning, and Computer Vision.

B. Step 2: Research Topic

Automated Multi-Class Skin Lesion Classification in Dermoscopic Imagery Using Deep Convolutional Neural Networks.

C. Step 3: Research Problem

Automated skin cancer classification models trained on standard benchmarks achieve high aggregate accuracy but suffer catastrophic diagnostic failure on life-threatening minority classes. This failure stems from severe dataset class imbalance and unweighted gradient loss optimization, causing models to favor benign majority samples at the expense of rare malignancies [8], [9].

D. Step 4: Literature Gap

While extensive literature reports overall accuracies exceeding 90% on the HAM10000 benchmark [10], published works consistently rely on naive random oversampling or downsampling [9]. Author-stated limitations in audited literature explicitly reveal:

- Jain et al. [9] state that image-space duplication causes severe overfitting and memorization of minority lesion artifacts.
- Nahata and Singh [12] state that deep networks are acutely prone to overfitting on classes with meager training samples.
- Dildar et al. [8] state that severe class imbalance across clinical benchmarks remains an unresolved barrier to real-world translation.

The precise unresolved gap: A rigorous, controlled comparative evaluation benchmarking cost-sensitive loss formulations (such as Focal Loss [13]) against standard Cross-Entropy across both heavyweight and edge-deployable CNN backbones under strictly leak-free, patient-deduplicated splits has not been systematically documented.

E. Step 5: Research Questions

- **RQ1:** How does the architectural choice among ResNet-50, DenseNet-121, EfficientNet-B0, and MobileNetV2 govern multi-class classification performance (Balanced Accuracy, Macro-F1) and computational efficiency (FLOPs, latency) on the HAM10000 benchmark under strictly identical training splits?
- **RQ2:** To what extent does the integration of class-balanced focal loss ($\gamma = 2.0$, α-weighted) improve diagnostic sensitivity on extreme minority categories (Dermatofibroma and Vascular lesions) relative to unweighted Categorical Cross-Entropy?

F. Step 6: Research Objectives

- **Objective 1:** To implement, train, and benchmark four standardized CNN backbones (ResNet-50, DenseNet-121, EfficientNet-B0, MobileNetV2) under identical preprocessing and 5-fold stratified cross-validation.
- **Objective 2:** To formulate and integrate a class-balanced focal loss objective function, evaluate per-class sensitivity across all seven diagnostic classes, and statistically analyze the performance shift on minority lesions.

G. Step 7: Target Benchmark Dataset

The investigation utilizes the publicly accessible, peer-reviewed HAM10000 (Human Against Machine with 10,000 training images) benchmark archive [10]. The dataset comprises 10,015 dermatoscopic images spanning seven diagnostic ground-truth categories confirmed via histopathology or expert consensus:

1) **nv (Melanocytic nevi):** 6,705 images (66.95% — Dominant majority).
2) **mel (Melanoma):** 1,113 images (11.11% — Lethal malignancy).
3) **bkl (Benign keratosis):** 1,099 images (10.97%).
4) **bcc (Basal cell carcinoma):** 514 images (5.13%).
5) **akiec (Actinic keratoses):** 327 images (3.26%).
6) **vasc (Vascular lesions):** 142 images (1.42% — Extreme minority).
7) **df (Dermatofibroma):** 115 images (1.15% — Extreme minority).

Data Leakage Prevention Protocol: The dataset contains multiple images corresponding to the same patient lesion. In accordance with rigorous methodology, data partitioning is grouped strictly by `lesion_id` ensuring no lesion appears simultaneously in train and test splits [10].

H. Step 8: Research Methodology

The experimental methodology comprises three controlled stages:

1) **Standardized Preprocessing:** All images are resized to 224×224 pixels and normalized using ImageNet channel mean ($\mu = [0.485, 0.456, 0.406]$) and standard deviation ($\sigma = [0.229, 0.224, 0.225]$). Online geometric augmentations (random horizontal/vertical flips, affine rotation $\pm 20^\circ$) are applied strictly during training.
2) **Network Optimization:** Models are initialized with ImageNet-1k pretrained weights [14]. Optimization utilizes AdamW with an initial learning rate $\eta = 10^{-4}$, weight decay of 10^{-2}, and cosine annealing scheduling across 50 epochs with batch size 32.
3) **Ablation Protocol:** Models are trained in parallel under two loss formulations:
   - Baseline: Standard Multi-Class Cross-Entropy Loss:

$$\mathcal{L}_{\text{CE}} = -\sum_{c=1}^{C} y_c \log(p_c) \tag{1}$$

   - Treatment: Class-Balanced Focal Loss [13]:

$$\mathcal{L}_{\text{FL}} = -\sum_{c=1}^{C} \alpha_c (1 - p_c)^{\gamma} \log(p_c) \tag{2}$$

   where $\gamma = 2.0$ acts as the focusing parameter and α_c is inversely proportional to class frequency N_c.

I. Step 9: Standardized Evaluation Metrics

To prevent misleading reporting, evaluation strictly avoids sole reliance on aggregate accuracy. The following formal metrics are logged:

- **Per-Class Sensitivity (Recall):**

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{3}$$

- **Per-Class Specificity:**

$$\text{Specificity}_c = \frac{TN_c}{TN_c + FP_c} \tag{4}$$

- **Macro-Averaged F1-Score:**

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^{C} \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{5}$$

- **Balanced Accuracy:** Arithmetic mean of per-class sensitivities.
- **Area Under the ROC Curve (Macro-AUC).**
- **Computational Footprint:** Total parameters, FLOPs, and mean inference latency (ms) across 1,000 test iterations on an NVIDIA RTX 4090 GPU and an ARM Cortex-A72 CPU.

J. Step 10: Expected Empirical Evidence and Falsification

- *To answer RQ1:* Tabulated confusion matrices and paired Wilcoxon signed-rank tests demonstrating whether DenseNet-121 or EfficientNet-B0 yields statistically significant macro-F1 improvements over MobileNetV2.
- *To answer RQ2:* Empirical sensitivity tables demonstrating whether Class-Balanced Focal Loss increases per-class recall on the `df` and `vasc` categories by more than 15 percentage points without reducing overall macro-specificity below 90%.
- *Status of Numerical Results:* All experimental evaluations are conducted under controlled live training; empirical outcomes will be populated following completed benchmark execution (`[RESULT TO BE PROVIDED]`).

XV. Research Question Quality Rubric

To enable student researchers to critically evaluate their own proposed research questions prior to submitting project proposals, Table VII establishes a 10-dimensional evaluation rubric scored on a 5-point Likert scale (1 = Unacceptable / Deficient, 5 = Exemplary / Publication-Ready).

XVI. Discussion and Threats to Validity

A rigorous research inquiry must proactively acknowledge and document threats to validity [3]:

- **Construct Validity:** Do the chosen metrics genuinely measure the intended phenomenon? In class-imbalanced medical imaging, overall accuracy suffers from catastrophic construct invalidity; researchers must mandate Macro-F1 and Balanced Accuracy.
- **Internal Validity:** Have all confounding variables been controlled? If Model A is trained with data augmentation while Model B is trained without it, any observed performance gap cannot be attributed to model architecture. Controlled benchmarking requires holding all preprocessing, batch sizes, optimizers, and random seeds strictly constant.
- **External Validity (Generalizability):** Do the empirical findings hold beyond the specific training benchmark? High accuracy on HAM10000 does not guarantee clinical efficacy on smartphone photography or diverse patient skin phototypes. Researchers must declare the operational boundary conditions of their work.
- **Conclusion Validity:** Are the statistical conclusions warranted? Researchers must avoid making sweeping superiority claims based on minor metric variations without establishing statistical significance through paired tests (e.g., Wilcoxon, McNemar) and multi-run dispersion reporting.

XVII. Conclusion

Identifying and formulating a valid research problem is the foundational milestone of Computer Science scholarship. This work has demonstrated that scientific research is not defined by the casual compilation of code or the uncritical application of established tools to standard data, but by the systematic, reproducible investigation of unresolved technical gaps.

By distinguishing broad topics from actionable problems, applying adversarial keyword triggers to academic literature, adhering to the 6-stage problem statement structure, and maintaining an unbroken logical chain connecting problems to questions, objectives, methodologies, and evidence, student scholars can conduct research that withstands rigorous peer review and contributes meaningful innovations to the computing discipline.

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TABLE I
COMPREHENSIVE TAXONOMY OF RESEARCH CLASSIFICATIONS IN COMPUTER SCIENCE

Research Type	Core Definition	Primary Purpose	Typical Methodology	Key Trade-offs	Computer Science Exemplar
A. Basic / Fundamental	Theoretical or mathematical inquiry driven by curiosity without immediate practical application.	Expand universal scientific laws and theoretical boundaries.	Formal proof, mathematical modeling, axiomatic derivation.	**Adv:** Durable universal foundations. **Lim:** No immediate commercial utility.	Proving lower bounds for matrix multiplication complexity or P vs. NP.
B. Applied	Systematic investigation designed to solve practical, real-world operational problems.	Produce practical interventions, deployable tools, and solutions.	System implementation, empirical benchmarking, optimization.	**Adv:** Direct societal/industrial impact. **Lim:** Context-specific; limited theoretical depth.	Developing lightweight quantized CNNs for dermatological triage on edge mobile devices.
C. Quantitative	Investigation relying on numerical measurements, statistical testing, and mathematical modeling.	Quantify variance, measure performance, establish statistical significance.	Controlled benchmarks, metric logging (F1, AUC), hypothesis testing.	**Adv:** High precision, replicability, statistical rigor. **Lim:** Obscures contextual and qualitative nuances.	Benchmarking inference latency and Top-1 accuracy across ResNet, DenseNet, and EfficientNet.
D. Qualitative	Non-numerical inquiry exploring subjective meanings, user mental models, and contextual dynamics.	Understand human experiences, developer reasoning, and usability.	Semi-structured interviews, thematic analysis, think-aloud protocols.	**Adv:** Deep contextual insight into human factors. **Lim:** Hard to generalize; prone to observer subjectivity.	Interviewing clinical dermatologists regarding their cognitive trust in saliency map explanations.
E. Mixed-Methods	Methodological integration combining both quantitative and qualitative data collection and analysis [2].	Provide comprehensive validation bridging numerical metrics and human rationale.	Convergent parallel design: benchmark logging paired with post-trial interviews.	**Adv:** Triangulates findings across diverse modalities. **Lim:** Resource-intensive; requires multi-paradigm skill.	Evaluating an AI diagnostic tool via classification metrics (quantitative) and clinician interviews (qualitative).
F. Experimental	Empirical strategy manipulating independent variables under controlled conditions to observe dependent variables [3].	Establish causal relationships and measure treatment effects.	Factorial designs, controlled A/B testing, randomized ablation studies.	**Adv:** Isolates pure causal relationships. **Lim:** Synthetic benchmark setup may diverge from messy reality.	Ablating loss formulations (Cross-Entropy vs. Focal Loss) on minority classes under identical training splits.
G. Exploratory	Initial research conducted into an ill-defined problem space lacking established precedent.	Clarify ambiguous concepts, generate preliminary hypotheses.	Literature scoping, pilot prototyping, unsupervised clustering.	**Adv:** Uncovers emerging paradigms and opportunities. **Lim:** Rarely provides definitive or conclusive proof.	Probing the vulnerability of multimodal foundation models against novel adversarial visual prompts.
H. Descriptive	Systematic characterization of the attributes, distributions, or phenomena of a system or population.	Provide accurate baseline profiles of current system states.	Observational monitoring, dataset profiling, architectural census.	**Adv:** Establishes rigorous baseline factual reference. **Lim:** Cannot establish causal mechanisms or 'why' phenomena occur.	Profiling class imbalance ratios and artifact prevalence across the HAM10000 and ISIC archives.
I. Explanatory	Rigorous inquiry establishing the causal mechanisms and functional dynamics governing system behavior.	Explain why specific phenomena, errors, or performance spikes emerge.	Mechanistic interpretability, gradient attribution, sensitivity analysis.	**Adv:** Unveils black-box internal mechanisms. **Lim:** High mathematical and diagnostic complexity.	Explaining why self-attention layers outperform convolution on dermoscopic lesion border irregularity.
J. Correlational	Statistical evaluation measuring the degree of association between two or more continuous variables.	Identify predictive relationships without manipulating variables.	Pearson/Spearman correlation coefficients, regression modeling.	**Adv:** Quantifies predictive associations on observational data. **Lim:** Correlation does not imply causation.	Measuring the statistical correlation between network parameter count and cross-dataset generalizability drop.
K. Case Study	In-depth empirical inquiry examining a contemporary phenomenon within its real-world context [6].	Understand real-world implementation barriers, workflows, and anomalies.	Single or multi-case field study, telemetry logging, artifact inspection.	**Adv:** Retains holistic real-world complexity. **Lim:** Limited statistical generalizability across domains.	Investigating the failure modes of deploying an AI diagnostic assistant in a rural hospital clinic.
L. Survey	Systematic collection of standardized data from a representative sample of individuals or artifacts.	Assess broad trends, developer practices, or configuration states.	Standardized questionnaires, automated GitHub code repository mining.	**Adv:** Wide coverage of developer practices and architectures. **Lim:** Response bias, self-reporting discrepancies.	Surveying 200 open-source PyTorch repositories to document transfer learning fine-tuning practices.
M. Action	Iterative, participatory inquiry combining practical system intervention with scientific reflection.	Solve an immediate operational pain point while advancing practice.	Iterative cycles: Diagnose $\rightarrow$ Plan $\rightarrow$ Act $\rightarrow$ Observe $\rightarrow$ Reflect.	**Adv:** Immediate operational improvement in the target setting. **Lim:** High risk of researcher advocacy	Iteratively redesigning a DevOps CI/CD automated testing pipeline with an engineering team.

TABLE II
SYSTEMATIC DEMARCATION OF RESEARCH HIERARCHY CONCEPTS

Research Concept	Epistemological Definition	Operational Function	Concrete Computer Science Exemplar
Research Area	A broad, overarching discipline or branch of scientific/technical knowledge.	Establishes the academic domain and community of practice.	Artificial Intelligence and Computer Vision.
Research Topic	A narrowed, focused subject matter within the broader area of inquiry.	Defines the specific computational or application domain.	Deep Learning-Based Dermoscopic Skin Lesion Classification.
Research Problem	An unresolved technical obstacle, theoretical dilemma, contradictory phenomenon, or operational failure in existing methods.	Identifies the core difficulty that requires systematic scientific resolution.	Standard deep convolutional neural networks exhibit severe classification performance degradation on rare malignant skin lesions due to extreme class imbalance and dominant-class optimization bias [8], [9].
Problem Statement	A formal, structured prose narrative articulating the context, reality, limitation, consequence, and necessity of investigation.	Conveys the scholarly justification and urgency of the inquiry to reviewers.	While automated deep learning models report high aggregate benchmark accuracy ($> 90\%$), clinical dermatological datasets such as HAM10000 exhibit acute class skew where benign nevi outnumber rare malignancies by over 60:1. Consequently, unweighted loss optimization causes standard CNNs to achieve abysmal sensitivity ($< 40\%$) on life-threatening minority classes such as dermatofibroma and vascular lesions. There is an urgent need to formulate class-balanced optimization frameworks to secure reliable diagnostic equity [9], [10].
Research Gap	A verified, documented void in current scientific literature where knowledge, evaluation, or methodology is lacking.	Identifies the exact scholarly space the current study will contribute to.	Despite widespread use of image augmentation, existing skin cancer studies fail to systematically benchmark cost-sensitive loss formulations (e.g., Focal Loss) against sampling techniques across lightweight edge backbones under identical preprocessing splits [9].
Research Question	A precise, interrogative, testable inquiry whose experimental answer addresses the research gap.	Guides the formulation of experimental protocols and hypothesis tests.	How does the incorporation of class-balanced focal loss impact the per-class recall and macro-F1 score of lightweight CNN architectures on minority classes in the HAM10000 benchmark compared to standard cross-entropy loss under identical training splits?
Research Objective	A concrete, measurable, action-oriented goal specifying the tasks executed to answer the research question.	Outlines the technical deliverables and milestones of the project.	To formulate, implement, and evaluate a class-balanced focal loss training strategy for MobileNetV2 and EfficientNet-B0 on the HAM10000 dataset, and measure the resulting recall improvement on minority lesion categories.
Hypothesis	A formal, testable, predictive proposition regarding the relationship between defined independent and dependent variables.	Provides a statistical baseline for empirical falsification and significance testing.	The application of class-balanced focal loss ($\gamma = 2.0$) will yield a statistically significant increase ($p < 0.05$) in minority-class recall ($> 15\%$ gain) compared to standard cross-entropy, without inducing a statistically significant drop in overall macro-accuracy.

TABLE III
KEYWORD TRIGGERS FOR DISCOVERING CANDIDATE RESEARCH PROBLEMS

Keyword Triggers	**Diagnostic Research Implication**
"However", "Despite", "Nevertheless"	Signals a pivotal contrast between an apparent achievement and an underlying unsolved limitation or constraint.
"Limitation", "Limitations", "Drawback"	Explicitly identifies the technical boundaries, unaddressed scenarios, or flaws acknowledged by the authors.
"Future Work", "Future Research"	Directly reveals the authors' roadmap of unexecuted experiments, extensions, and open problems.
"Challenge", "Remains an open problem"	Flags a recognized computational or theoretical barrier that the current state of the art has failed to overcome.
"Generalization", "Domain shift"	Indicates that the model was tested under isolated, single-center conditions and may fail in wild environments.
"Computational cost", "Latency"	Indicates the proposed model is excessively large, slow, or resource-heavy for real-time mobile triage.

TABLE IV
CRITICAL COMPARATIVE ANALYSIS OF DEFICIENT VS. METHODOLOGICALLY ROBUST RESEARCH QUESTIONS

Novice / Flawed Formulation	Rigorous Academic Formulation	Methodological Rationale and Comparative Critique
Question 1 (AI Fundamentals): *"What is artificial intelligence?"*	*"How does transfer learning from ImageNet affect multi-class skin lesion classification performance under limited sample sizes compared to training from random initialization?"*	**Defect:** The flawed question is an elementary textbook query with no research gap, zero hypothesis, and no empirical boundary. **Rigorous Merit:** Bounds the inquiry to transfer learning, defines independent variables (pretrained vs. random weights, sample size), dependent variables (multi-class performance), and the target domain (skin lesions).
Question 2 (Comparative ML): *"Which is the best machine learning algorithm?"*	*"How does Random Forest compare with Support Vector Machines in classification accuracy, macro-F1, and training latency on high-dimensional genomic feature data under identical 10-fold cross-validation?"*	**Defect:** "Best" is an unscientific, subjective superlative with no meaning in computer science (No-Free-Lunch Theorem). **Rigorous Merit:** Specifies exact algorithmic baselines, explicit evaluation metrics (accuracy, F1, latency), concrete data modality (high-dimensional genomics), and a leak-free validation protocol.
Question 3 (Cybersecurity Anomaly Detection): *"Can AI stop cyber attacks?"*	*"To what extent does an attention-augmented LSTM autoencoder reduce false positive alarm rates in network intrusion detection on the CIC-IDS2017 benchmark compared to standard Isolation Forests?"*	**Defect:** A binary "yes/no" marketing claim that is impossible to experimentally falsify or scientifically defend. **Rigorous Merit:** Replaces sensational claims with a measurable objective (false positive reduction), defines exact architectures, baselines, and a standard public benchmark.
Question 4 (Cloud / Systems Engineering): *"How fast is cloud computing?"*	*"What is the impact of function memory allocation tiering (128 MB to 3072 MB) on the 99th-percentile tail latency of cold starts in serverless microservice architectures?"*	**Defect:** Vague, undefined scope without operational parameters, target platforms, or measurable metrics. **Rigorous Merit:** Isolates specific system variables (memory allocation tiers, cold starts), targets a precise performance metric ($p99$ tail latency), and bounds the architectural context (serverless).

TABLE V
TAXONOMY OF ACTION VERBS FOR COMPUTER SCIENCE RESEARCH OBJECTIVES

Category	Approved Measurable Verbs
Investigation & Analysis	*Investigate, Analyze, Compare, Benchmark, Characterize, Quantify, Profile, Dissect.*
Design & Engineering	*Design, Formulate, Architect, Construct, Synthesize, Derive, Implement, Quantize.*
Evaluation & Validation	*Evaluate, Assess, Validate, Measure, Verify, Falsify, Ablate, Benchmark.*
Optimization & Tuning	*Optimize, Calibrate, Minimize, Maximize, Regularize, Accelerate, Prune.*
Prohibited Vague Verbs	**Methodological Defect**
"Understand", "Know"	Internal cognitive states; impossible to verify, test, or experimentally measure.
"Learn", "Study"	Educational activities, not scientific contributions to human knowledge.
"Discuss", "Explore"	Conversational actions; lack operational deliverables unless strictly bounded by empirical protocols.

TABLE VI
ONE-TO-ONE ALIGNMENT OF RESEARCH QUESTIONS TO RESEARCH OBJECTIVES

Research Question (RQ)	Direct Research Objective
RQ1 (Inductive Bias): How do convolutional architectures (ResNet-50, DenseNet-121, EfficientNet-B0, MobileNetV2) compare in diagnostic macro-F1 and inference latency under identical preprocessing splits?	**Objective 1:** To systematically benchmark and analyze the classification performance, memory footprint, and inference latency of the four candidate CNN architectures under strictly identical training, validation, and testing protocols.
RQ2 (Class Imbalance): To what extent does class-balanced focal loss improve minority-class recall compared to unweighted cross-entropy on the HAM10000 benchmark?	**Objective 2:** To formulate, implement, and evaluate a class-balanced focal loss training strategy and quantify the resulting per-class recall gains on dermatofibroma and vascular lesions.
RQ3 (Transfer Learning Dynamics): How does layer freezing depth affect classification accuracy and training time during fine-tuning?	**Objective 3:** To design and ablate a three-tier layer-freezing protocol to determine the optimal Pareto trade-off between diagnostic accuracy and training convergence speed.

TABLE VII
THE 10-DIMENSIONAL RESEARCH QUESTION QUALITY EVALUATION RUBRIC

Evaluation Dimension	Deficient Formulation (Score: 1)	Marginal Formulation (Score: 3)	Exemplary Formulation (Score: 5)
1. Specificity	Vague, boundless concept without scope limits.	Mentions a general algorithm and task.	Explicitly specifies models, datasets, hyperparameter regimes, and conditions.
2. Researchability	Subjective, philosophical, or unfalsifiable assertion.	Researchable, but requires subjective human interpretation.	Fully falsifiable through empirical data collection and statistical testing.
3. Technical Feasibility	Demands supercomputing clusters or non-existent data.	Feasible, but requires borderline GPU resources or commercial APIs.	Solvable within standard academic GPU hardware and open-access benchmarks.
4. Literature Gap Anchoring	Re-asks a solved question or provides no citations.	Mentions a known topic but without documented gap evidence.	Directly cites explicit author limitations from peer-reviewed literature.
5. Data Availability	Relies on proprietary, private, or unreleased data.	Data exists, but requires extensive undocumented cleaning.	Verified open-access benchmark with documented splits and ground truth.
6. Experimental Tractability	Cannot be executed via software or algorithmic testing.	Requires complex hybrid setups with ill-defined variables.	Controlled A/B or factorial experimental design with isolated variables.
7. Methodological Alignment	Methodology cannot answer the proposed inquiry.	Methodology addresses the general area but misses key variables.	Experimental methodology is designed specifically to produce the exact answer.
8. Evidentiary Decisiveness	Results cannot provide clear statistical evidence.	Results provide indirect or ambiguous indications.	Produces definitive quantitative metrics with confidence intervals.
9. Discipline Relevance	Pure management, marketing, or general science inquiry.	Applied computing, but lacks technical depth or innovation.	Directly advances Computer Science, algorithms, systems, or data engineering.
10. Scope Calibration	Overly broad ("solve cancer") or overly trivial ("change color").	Slightly too broad for an undergraduate project timeline.	Calibrated perfectly for rigorous semester-long execution and publication.